

ANGELOPOULOS & BATES (2022) – A GENTLE INTRODUCTION TO CP

Foundations and Trends in ML (arXiv:2107.07511). De-facto practitioner reference.

- Thesis: any black-box $\hat{f}$ + an exchangeable calibration set + a chosen nonconformity score $s(x, y)$ yields finite-sample valid prediction sets, no distributional assumptions.
- Three layers: recipe (4 steps) + score catalogue (APS, RAPS, CQR, conformal Bayes, ...) + extension survey (risk control, weighted CP, Jackknife+, group-conditional, ...).
- Every method ships with a Python notebook — the engineering hook that drove adoption.
- Authors: Anastasios Angelopoulos (UC Berkeley), Stephen Bates (MIT).

ANGELOPOULOS–BATES: THE FOUR-STEP RECIPE IN FORMULAS

1. Heuristic uncertainty: any predictor $\hat{f}$.
2. Nonconformity score $s(x, y) \in \mathbb{R}$.
3. Calibration quantile:

$$\underbrace{\hat{q}}_{\text{cutoff}} = \text{Quantile}\left(\underbrace{\{s_i\}_{i=1}^n}_{\text{cal. scores}} ; \underbrace{\frac{\lceil (n+1)(1-\alpha) \rceil}{n}}_{\substack{\text{slightly above } 1-\alpha; \\ +1 \text{ corrects for test point}}} \right).$$

4. Prediction set $\mathcal{C}(X_{n+1}) = \{y : s(X_{n+1}, y) \leq \hat{q}\}$.

Score design is the user's choice; the coverage guarantee survives any choice.

ANGELOPOULOS–BATES: COVERAGE THEOREM + BETA COROLLARY

Theorem (Marginal coverage)

$$(X_i, Y_i)_{i=1}^{n+1} \text{ exchangeable} \Rightarrow \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \in [1-\alpha, 1-\alpha + \frac{1}{n+1}].$$

- Realised coverage on a *fixed* calibration set:

$$\underbrace{C}_{\text{realised coverage}} \sim \text{Beta}\left(\underbrace{\lceil (n+1)(1-\alpha) \rceil}_{\text{shape 1}}, \underbrace{n+1 - \lceil (n+1)(1-\alpha) \rceil}_{\text{shape 2}}\right).$$

- Rule of thumb:** $n \approx 1,000$ keeps realised coverage within $\pm 1\%$ of nominal.
- Diagnostics:** size-stratified and feature-stratified coverage as proxies for X-conditional validity.

ANGELOPOULOS–BATES: SCORE CATALOGUE AT A GLANCE

- Classification: $s = 1 - \hat{f}_y(x)$ (fixed sets) / APS (ranked cumulative softmax, **input-difficulty adaptive**) / RAPS (regularised APS, **prevents pathological tail sets**).
- Regression: $s = |y - \hat{f}(x)|$ (constant width) / $s = |y - \hat{f}(x)|/\hat{\sigma}(x)$ (**σ -scaled**) / CQR:

$$s(x, y) = \max(\hat{q}_{\alpha/2}(x) - y, y - \hat{q}_{1-\alpha/2}(x)).$$

- Bayes: $s = -\log p_{\theta}(y | x)$ (**conformalises any posterior predictive**).
- Distributional: $s = \text{PIT score}$ (intervals from any functional of the predictive distribution).
- All four: **same theorem**, different set shapes.

ANGELOPOULOS–BATES: BEYOND COVERAGE + CONNECTIONS

- **Conformal Risk Control**: replace coverage with any **monotone** loss $L(\lambda) \in [0, 1]$; tune λ so $\mathbb{E}[L] \leq \alpha$. Generalises FNR, FDR, mask coverage.
- **Learn-Then-Test**: extends risk control to **non-monotone** risks via multiple testing on a grid of λ .
- **Approximate conditional coverage**: class-conditional, group-balanced, Mondrian CP.
- **Compute–statistics spectrum**: Full CP (exact, $O(|\mathcal{Y}|)$ refits) $\rightarrow$ Cross/CV+ (K -fold) $\rightarrow$ Jackknife+ ($1-2\alpha$) $\rightarrow$ Split CP (one fit, loses cal. data).
- **Reading prerequisite for** Stocker (time-series companion) and Dieuleveut–Zaffran (advanced tutorial).

DIEULEVEUT & ZAFFRAN (2025) – HI! PARIS TUTORIAL (91 SLIDES)

Joint UAI/ICML tutorial, delivered Hi! PARIS Summer School (École Polytechnique, July 2025).

Cleanest entry to the Zaffran research line.

- Pedagogical arc: motivation → SCP regression+classification → CQR/APS → coverage hierarchy → Full CP & Jackknife+ → breaking exchangeability → applied case studies.
- Distinguishing feature vs. Angelopoulos–Bates: explicit mathematical treatment of when CP coverage really holds — conditional impossibility, PAC bounds, online-update guarantees.
- Pairs with Zaffran PhD (2024) for full proofs and Angelopoulos–Bates for the i.i.d. recipe.

DIEULEVEUT–ZAFFRAN: THE QUANTILE LEMMA AS WORKHORSE

Lemma (Quantile lemma, paraphrased)

$S_1, \dots, S_{n+1}$ exchangeable $\Rightarrow \mathbb{P}(S_{n+1} \leq \hat{q}_\beta) \geq \beta$ where $\hat{q}_\beta$ is the $\lceil (n+1)\beta \rceil$ -th smallest of $S_1, \dots, S_n$.

- Single technical workhorse behind every CP variant.
- Apply to SCP scores $\rightarrow$ split conformal coverage.
- Apply to augmented scores $\rightarrow$ full conformal coverage.
- Apply to LOO residuals + comparison-matrix $\rightarrow$ Jackknife+ $\geq 1 - 2\alpha$.
- Apply to reweighted scores $\rightarrow$ weighted CP under covariate shift.

Once you see the lemma, every variant looks like a corollary, not a separate invention.

DIEULEVEUT-ZAFFRAN: THE COVERAGE HIERARCHY (WEAKEST TO STRONGEST)

1. **Marginal:** $\mathbb{P}(Y \in \hat{C}(X)) \geq 1 - \alpha$ (over joint draw of test *and* cal. set).
2. **Calibration-conditional** (Vovk 2012, Bian-Barber 2023): for fixed cal. set,
 $\mathbb{P}_{\text{test}}(Y \in \hat{C}(X)) \geq 1 - \alpha - \varepsilon(n, \delta)$ w.p. $\geq 1 - \delta$.
3. **X-conditional:** $\mathbb{P}(Y \in \hat{C}(X) \mid X = x) \geq 1 - \alpha \forall x$.

Lemma (Vovk impossibility, paraphrased)

Distribution-free X-conditional coverage at non-trivial level $\Rightarrow$ sets of infinite Lebesgue measure on positive-probability subsets.

Approximate substitutes: class-/group-conditional CP, local adaptivity (CQR), asymptotic conditional via consistent QR.

DIEULEVEUT–ZAFFRAN: BREAKING EXCHANGEABILITY + JACKKNIFE+

- **Jackknife+** (Barber–Candès–Ramdas–Tibshirani 2021):

$$\mathbb{P}\left(Y_{n+1} \in \mathcal{C}^{\text{J}^+}(X_{n+1})\right) \geq 1 - 2\alpha \quad \underbrace{\text{(distribution-free, no split)}}_{\text{LOO + comparison-matrix argument}}$$

- **Known covariate shift** (Tibshirani et al. 2019): reweight scores by $w(x) = dQ_X(x)/dP_X(x)$; marginal coverage on Q at cost of effective-sample-size penalty.
- **Time series, online** (Gibbs–Candès 2021): **ACI** updates α_t via gradient rule; long-run miscoverage $\rightarrow \alpha$ for *any* sequence.

DIEULEVEUT–ZAFFRAN: APPLIED CASE STUDIES + FRONTIER

- **Healthcare image-to-image regression** (Angelopoulos et al. 2022): conformal masks around pixel-wise predictions for medical imaging; per-pixel coverage gives clinicians defensible uncertainty.
- **French day-ahead electricity prices**: ACI handles spike-driven distribution drift better than rolling SCP; 2021 European energy crisis as in-sample stress test.
- **Recent Dieuleveut-group results**:
 - **CP-MDA-Nested** (Zaffran–Josse–Romano–Dieuleveut 2024): mask-conditional validity under MCAR/MAR missingness.
 - **Valid Selection among Conformal Sets**: test-time set choice preserving coverage.
- **Outlook**: selective inference + FDR control, beyond-indicator risk control, aggregation of predictive sets (Gasparin–Ramdas).

STOCKER, MALGORZEWICZ, FONTANA, BEN TAIEB (2025) – CP FOR TIME SERIES

arXiv:2511.13608 — the **time-series companion** to Angelopoulos–Bates.

- Diagnosis: time series violates **exchangeability** via two independent mechanisms:
 1. **Temporal dependence** (ARMA-style filtration); observations carry info through time.
 2. **Distribution shift** (abrupt breaks, gradual drift, seasonality).
- **Headline contribution**: a four-family mechanistic taxonomy of all modern CP-for-time-series methods.
- Self-contained re-derivation of finite-sample marginal + conditional SCP coverage under β -mixing.

STOCKER ET AL.: SCP UNDER β -MIXING (THEOREM A.3.1)

Theorem (Informal, paraphrased)

Under stationarity + β -mixing of (X_t, Y_t) (Assumptions A1–A5),

$$\underbrace{\left| \mathbb{P}(Y_t \in \hat{C}_t(X_t)) - (1 - \alpha) \right|}_{\substack{\text{coverage gap of} \\ \text{vanilla SCP at index } t}} \rightarrow 0 \quad \text{as the calibration window grows.}$$

Similar result for X -conditional coverage under additional assumptions.

- **Practical implication:** on stationary β -mixing data, **vanilla SCP still works**. No fancy methods needed.
- Cleaner version of Oliveira et al. (2024); full proofs in appendix.
- Justifies SCP as the default for forecasting in stable regimes (stationary ARMA, many GARCH variants).

STOCKER ET AL.: THE FOUR-FAMILY TAXONOMY

Every CP-for-time-series method modifies **exactly one** object of the SCP procedure:

1. **Reweight calibration data** ($\rightarrow$ **WCP**, **NexCP**): recency weights or learned weights.
2. **Refresh residual pool via OOB bootstrap** ($\rightarrow$ **EnbPI**, **SPCI**): bootstrap LOO + sliding window.
3. **Adapt α_t online** ($\rightarrow$ **ACI**, **AgACI**, **Conformal PID**): gradient update on target miscoverage.
4. **Randomise over blocks** ($\rightarrow$ **Block CP**): block-level exchangeability replaces point-level.

Four objects, four families. Hybrid methods compose families.

STOCKER ET AL.: SIMULATION FINDINGS

Controlled study: SCP, WCP×3, ACI×3, EnbPI×3, Block-SCP on AR(1), ARMA(1,1), mean-shift, GARCH(1,1).

- **Stationary β -mixing** (AR, ARMA): **vanilla SCP is valid and efficient**. WCP, ACI, EnbPI match coverage with no gain.
- **Abrupt mean shift**: SCP, Block-SCP, WCP-window **fail to cover**. ACI, EnbPI, WCP-exp, WCP-linear **self-correct**.
- **Non-obvious failure**: Block-SCP under-covers *even on stationary β -mixing data* — counter-intuitive cost of randomising blocks.

STOCKER ET AL.: PRACTITIONER DEFAULTS + FRONTIER

Defaults from the paper's discussion:

- Stationary → **SCP**.
- Cheap shift-robustness → **WCP-exp** or **WCP-linear**.
- Active feedback / unknown drift → **ACI** (or AgACI).
- Heteroskedasticity matters, compute available → **EnbPI** / **SPCI**.

Research frontier:

- Hybrid methods (ACI + SPCI conditional quantiles, learned WCP weights).
- Distributional conformal forecasting.
- Multivariate / functional time series, multi-step horizons.
- Long-memory processes (where β -mixing fails).

Code reproducibility: accompanying GitHub repo by Malgorzewicz.

XU & XIE (2023) – ENBPI (IEEE TPAMI)

arXiv:2010.09107. The canonical “refresh residual pool” method in Stocker’s taxonomy.

- Problem: time series breaks exchangeability → naive SCP loses guarantees; ARIMA needs parametric assumptions; quantile regression needs heavy retraining.
- **Core insight:** a **bootstrap ensemble** already contains LOO information — for each i , the bootstrap models that didn’t see i aggregate to a leave-one-out estimator $\hat{f}^{-i}$ **for free**.
- Result: **no data split**, **no retraining at test time**, and **asymptotic conditional coverage** under β -mixing.
- Authors: Chen Xu (Bloomberg, ex-Georgia Tech), Yao Xie (Georgia Tech). Follow-up: SPCI (2022).

ENBPI: THE BOOTSTRAP-LOO ALGORITHM

Training $(X_t, Y_t)_{t=1}^T$, ensemble size B :

1. Train B bootstrap predictors $\{\hat{f}^{(b)}\}_{b=1}^B$ on resampled training sets.
2. For each t , aggregate **only** the bootstrap models that didn't see t :

$$\underbrace{\hat{f}^{-t}(x)}_{\text{LOO estimator at index } t} = \frac{1}{|\{b : t \notin \mathcal{I}_b\}|} \sum_{b: t \notin \mathcal{I}_b} \hat{f}^{(b)}(x).$$

3. LOO residuals $\hat{\varepsilon}_t = Y_t - \hat{f}^{-t}(X_t)$.
4. **Test-time interval**: sliding-window quantile of $\{\hat{\varepsilon}\}$; asymmetric via inner optimisation over $\beta \in [0, \alpha]$ to minimise width.
5. As new Y_t arrives, append to residual window. **$\hat{f}$ never retrained.**

ENBPI: COVERAGE THEOREM + COST ANALYSIS

Theorem (Asymptotic coverage gap, informal)

Under β -mixing of (X_t, Y_t) and consistency of the bootstrap LOO estimator,

$$\underbrace{\left| \mathbb{P}(Y_{T+h} \in \hat{C}_h(X_{T+h})) - (1 - \alpha) \right|}_{\substack{\text{coverage gap} \\ \text{at horizon } h}} \xrightarrow{T \rightarrow \infty} 0.$$

*Holds for **both marginal and X-conditional** coverage. Bounds also given on interval-width gap vs. oracle.*

- **Cost:** one ensemble fit upfront, $O(1)$ per test point thereafter.
- **Conditional guarantee** is rare for CP-on-time-series — usually impossible at finite n .
- Price: needs consistency of $\hat{f}^{-t}$ (a non-trivial assumption that gives up distribution-freeness).

ENBPI: EMPIRICAL BENCHMARK

Evaluated on five high-stakes domains:

- Solar/wind energy generation (renewable supply forecasting).
- Multivariate anomaly detection (intrusion / fault).
- Greenhouse-gas concentration (environmental monitoring).
- Appliance energy consumption (smart grid).
- Beijing air quality (PM2.5 forecasting under regime shifts).

Beats: ARIMA, ICP, WeightedICP, QOOB, AdaptCI (Gibbs–Candès), J+aB.

- Shorter intervals at nominal coverage, robust to missing data, handles change points via sliding-window length.

ENBPI: LIMITATIONS + CONNECTIONS

- Degrades as β -mixing slows toward long memory.
- For **abrupt shift**, pair with ACI (online α -update). Hybrid: ACI + EnbPI.
- For **heteroskedasticity**, use SPCI follow-up (learned QR score).
- **Lineage:**
 - Predecessor: Kim–Xu–Barber 2020 J+aB (**exchangeable** bootstrap LOO).
 - Successor: SPCI 2022 (**learned quantile regressor** as score).
 - Causal extension: DR-ACI (Koukorinis 2026) for treatment-effect estimation under temporal dependence + DML.
- Slots into Stocker's taxonomy as the headline "**residual refresh**" method.

BAO, COLOMBO, MANOKHIN, CAO, LUO (2025) – CONFORMAL REGRESSION REVIEW

Proc. Conformal and Probabilistic Prediction and Applications (COPA), PMLR 266:1–23. **The practitioner decision aid** for univariate regression.

- Problem: regression branch of CP literature has **inconsistent notation** and a **proliferation of methods** (SCP, CQR, CHR, CTI, DCP, normalised SCP...) with no head-to-head benchmark.
- Contribution: **unified notation** + **categorisation by score family** + **empirical benchmark** on 12 real-world datasets.
- Stays focused on **univariate continuous targets** — deliberately narrow scope keeps comparisons apples-to-apples.
- Pairs with Angelopoulos–Bates for foundations → Bao tells you *which* regression score to pick.

BAO ET AL.: THE 8 METHODS ORGANISED BY SCORE FAMILY

- **Residual-based (symmetric):**
 - **SCP** (Papadopoulos 2002, Lei et al. 2018): $s = |y - \hat{f}(x)|$.
 - **Normalised SCP**: $s = |y - \hat{f}(x)| / \hat{\sigma}(x)$.
- **Quantile-based (asymmetric, locally adaptive):**
 - **CQR** (Romano et al. 2019): $s = \max(\hat{q}_{\alpha/2} - y, y - \hat{q}_{1-\alpha/2})$.
 - **CHR** (Sesia-Romano 2021): histogram of conditional density.
 - **CTI**: thresholded conformal sets from full distributional output.
- **Distributional (full predictive CDF):**
 - **DCP** (Chernozhukov et al. 2021): PIT-based score.
- Plus locally-weighted-residual variants and conformal predictive systems.

BAO ET AL.: SIMULATION INSIGHTS

- **Homoscedastic data:** symmetric (SCP) and asymmetric (CQR, CHR) methods all produce similar widths. SCP is simplest and competitive.
- **Heteroscedastic data:** SCP gives **uniform-width bands** (too wide low-var, too narrow high-var). Normalised SCP and CQR **track the conditional spread**, giving local widths.
- **Multi-modal $Y | X$:** SCP and CQR cover both modes **plus the gap between them** (wasteful). CHR and DCP can produce **disjoint sets** excluding the low-density region — dramatically narrower at the same coverage.

BAO ET AL.: EMPIRICAL FINDINGS + TRADE-OFF TRIANGLE

- No single winner across 12 datasets — ranking depends on data characteristics.
- Marginal coverage satisfied by all methods (theorem); they differ on interval width, conditional calibration, compute cost.
- CQR / CHR / CTI dominate on heteroscedastic / multi-modal data.
- SCP / Normalised SCP competitive on roughly homoscedastic data; lowest compute and tuning footprint.
- DCP produces best conditional calibration but costs extra distributional regressor + more calibration data.

Trade-off triangle: computational cost $\leftrightarrow$ interval width $\leftrightarrow$ calibration sample size.

BAO ET AL.: METHOD-SELECTION HEURISTIC

- **Default:** **Normalised SCP** if $\hat{\sigma}(x)$ available; **plain SCP** otherwise. Both have lowest compute, finite-sample marginal coverage.
- **Suspected heteroscedasticity** (financial returns, energy, biomedical): **CQR**.
- **Multi-modal $Y \mid X$ or downstream needs full distribution:** **CHR** or **DCP**.
- **Diagnostic use:** run two or three methods; substantially different shapes signal underlying conditional structure (different widths $\rightarrow$ heteroscedasticity; different shapes $\rightarrow$ multi-modality).
- **Pairs with:** Angelopoulos–Bates for the recipe (read first), then Bao for per-variant trade-offs, then the primary reference of whichever method you select.